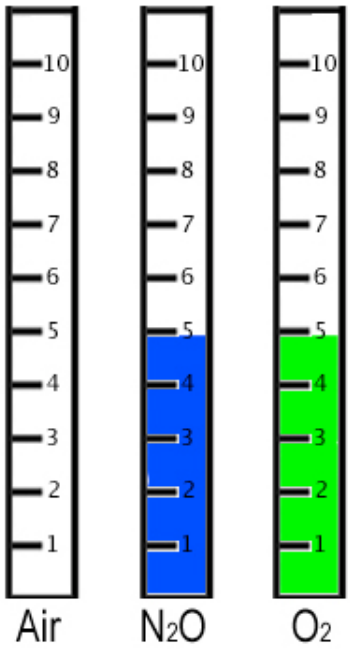
When you turn the flowmeter dials on the anesthesia machine, you change the mixture of gases being delivered to the patient. If you are only using oxygen, then you are delivering 100% oxygen no matter what flow rate in liters/minute you choose. It can begin to seem a little more complicated, however, when you add multiple gases as shown in the example to the right.

The concentration of oxygen (FiO_2) is expressed as a percent and is often written as a decimal. For example, an FiO_2 of 50% and an FiO_2 of 0.5 represent the same concentration of oxygen.



Example #1



Look at the example to the left. The flowmeter indicates that the oxygen knob is turned to 5 L/min and the nitrous oxide knob is also turned to 5 L/min.

Because the FiO_2 is the fraction of oxygen being delivered, our formula is a fraction as illustrated below:

$$FiO_2 = \frac{\text{L/min } O_2}{(\text{L/min } O_2 + \text{L/min } N_2O)}$$

Or

$$FiO_2 = \frac{5 \text{ L/min O2}}{(5 \text{ L/min } O_2 + 5 \text{ L/min } N_2O)} = \frac{5}{10}$$

Or

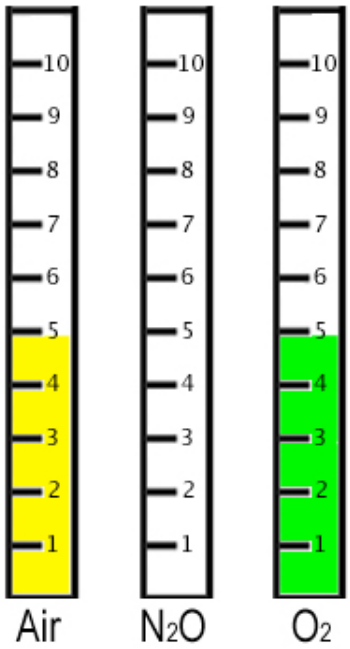
FiO_2 = 0.5 or 50%

For the purposes of this workbook, we will be rounding our answers to the nearest whole percentage point.

The example above is easy for anyone to calculate. Just divide the liters/minute of oxygen flow by the total liters/minute of all gases combined. But, what if you turn on the Air dial? Introducing air adds a slight complication because only 21% of air is oxygen. But, by adding one more factor to our formula, we can easily calculate the correct answer.

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Example #2



In this example, we have two gases turned on—oxygen and air. We'll upgrade our formula to account for all three gases.

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

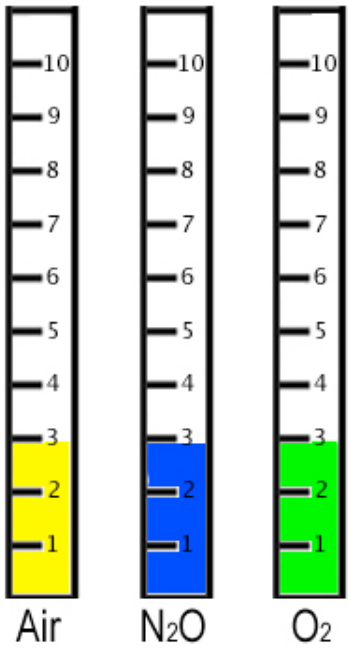
$$FiO_2 = \frac{5\ L/min\ O_2 + (0.21)(5\ L/min\ Air)}{(5\ L/min\ O_2 + 0\ L/min\ N_2O + 5\ L/min\ Air)}$$

Or

$$FiO_2 = \frac{5 + 1.05}{10}$$

Or $FiO_2 = 0.605$ or 61%

Example #3



In this example, all three gases are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

$$FiO_2 = \frac{3\ L/min\ O_2 + (0.21)(3\ L/min\ Air)}{(3\ L/min\ O_2 + 3\ L/min\ N_2O + 3\ L/min\ Air)}$$

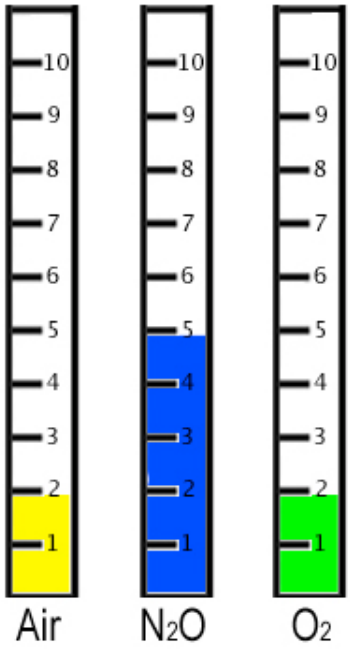
Or

$$FiO_2 = \frac{3 + 0.63}{9}$$

Or $FiO_2 = 0.403$ or 40%

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Example #4



In this example, all three gases are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

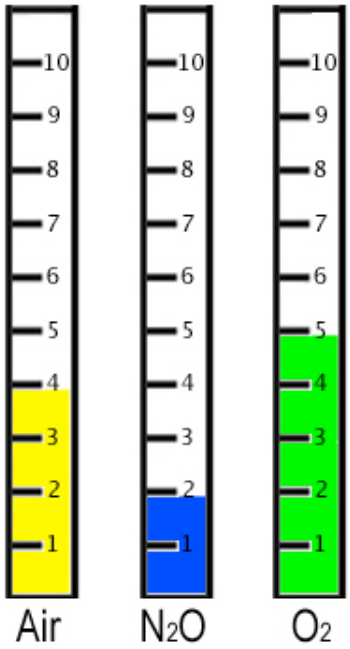
$$FiO_2 = \frac{2\ L/min\ O_2 + (0.21)(2\ L/min\ Air)}{(2\ L/min\ O_2 + 5\ L/min\ N_2O + 2\ L/min\ Air)}$$

Or

$$FiO_2 = \frac{2 + 0.42}{9}$$

Or $FiO_2 = 0.2688$ or 27%

Example #5



In this example, all three gases are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

$$FiO_2 = \frac{5\ L/min\ O_2 + (0.21)(4\ L/min\ Air)}{(5\ L/min\ O_2 + 2\ L/min\ N_2O + 4\ L/min\ Air)}$$

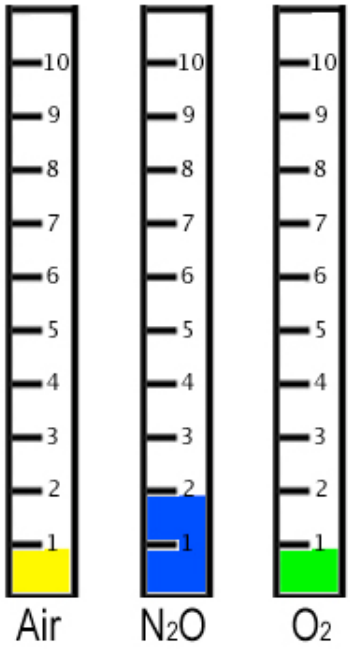
Or

$$FiO_2 = \frac{5 + 0.84}{11}$$

Or $FiO_2 = 0.5309$ or 53%

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Example #6



In this example, all three gases are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

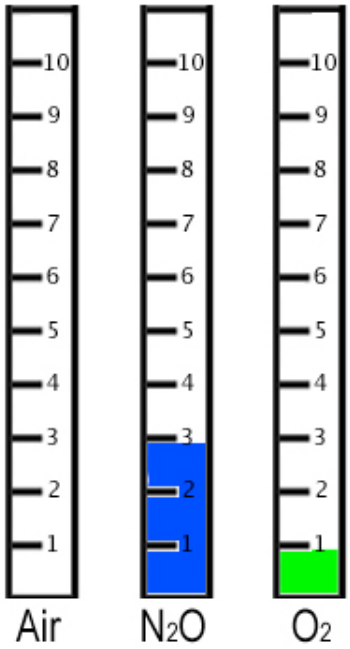
$$FiO_2 = \frac{1\ L/min\ O_2 + (0.21)(1\ L/min\ Air)}{(1\ L/min\ O_2 + 2\ L/min\ N_2O + 1\ L/min\ Air)}$$

Or

$$FiO_2 = \frac{1 + 0.21}{4}$$

Or FiO_2 = 0.3025 or 30%

Example #7



In this example, only nitrous oxide and oxygen are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

$$FiO_2 = \frac{1\ L/min\ O_2 + (0.21)(0\ L/min\ Air)}{(1\ L/min\ O_2 + 3\ L/min\ N_2O + 0\ L/min\ Air)}$$

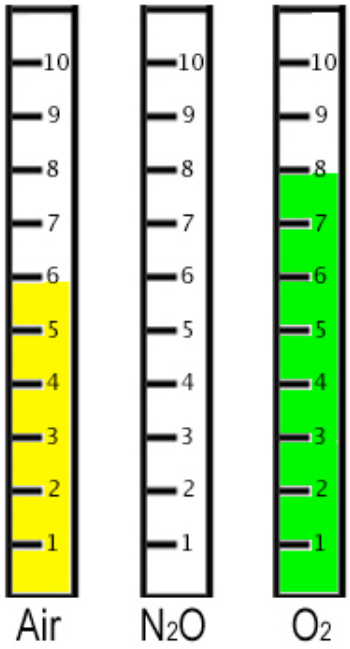
Or

$$FiO_2 = \frac{1}{4}$$

Or FiO_2 = 0.25 or 25%

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Example #8



In this example, only air and oxygen are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

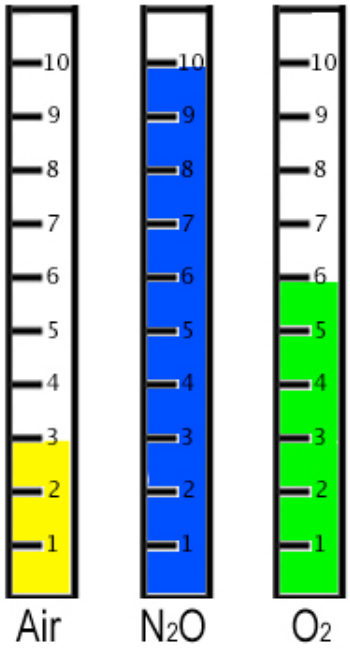
$$FiO_2 = \frac{8\ L/min\ O_2 + (0.21)(6\ L/min\ Air)}{(8\ L/min\ O_2 + 0\ L/min\ N_2O + 6\ L/min\ Air)}$$

Or

$$FiO_2 = \frac{8 + 1.26}{14}$$

Or $FiO_2 = 0.661$ or 66%

Example #9



In this example, all three gases are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

$$FiO_2 = \frac{6\ L/min\ O_2 + (0.21)(3\ L/min\ Air)}{(6\ L/min\ O_2 + 10\ L/min\ N_2O + 3\ L/min\ Air)}$$

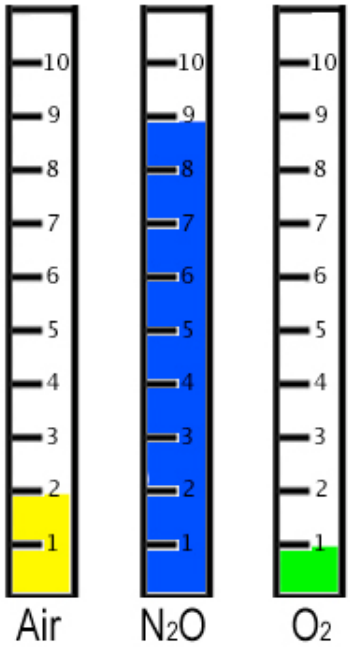
Or

$$FiO_2 = \frac{6 + 0.63}{19}$$

Or $FiO_2 = 0.3489$ or 35%

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Example #10



In this example, all three gases are turned on. Set up the problem using the formula:

$$FiO_2 = \frac{L/min\ O_2 + (0.21)(L/min\ Air)}{(L/min\ O_2 + L/min\ N_2O + L/min\ Air)}$$

Or

$$FiO_2 = \frac{1\ L/min\ O_2 + (0.21)(2\ L/min\ Air)}{(1\ L/min\ O_2 + 9\ L/min\ N_2O + 2\ L/min\ Air)}$$

Or

$$FiO_2 = \frac{1 + 0.42}{12}$$

Or $FiO_2 = 0.118$ or 12%

Our final example illustrates a specific point about the nitrous and oxygen flowmeters. Specifically, the configuration shown is not possible with an anesthesia machine. To prevent the delivery of a hypoxic mixture of oxygen (less than 21%), the nitrous oxide and oxygen dials are linked together. If you raise the nitrous oxide dial, it will automatically turn the oxygen dial so that a potentially fatal mixture like this example cannot be delivered to the patient.

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